Thakaa at NADI 2026 Shared Task: ROVER Ensembling and Multi-Representation Fusion for Robust Arabic ASR and Dialect Identification

Abdulaziz Otaif, Hazem Bakhshwain, Abdulhamid Aldoobi,
Meshal Alamr, Abdullah Aldahlawi
Thakaa, Advanced Company for Information Technology
Riyadh, Saudi Arabia
{a.otaif,h.bakhshwain,a.aldoobi,m.alamr,aldahlawi}@thakaait.net

Abstract

We describe Thakaa's systems for NADI 2026 subtasks 1.1 and 1.3 and Task 2, covering noisy country-level Arabic automatic speech recognition (ASR), Tunisian code-switched ASR, and spoken dialect identification. The ASR systems combine heterogeneous connectionist temporal classification (CTC), encoder–decoder, and audio-LLM recognizers through ROVER, with country-aware routing for subtask 1.1 and a frame-rate-corrected CTC anchor for subtask 1.3. The dialect identifier fuses Ara-BEST-RQ acoustic scores with lexical evidence extracted from Cohere and Whisper transcripts by a partially fine-tuned MARBERTv2 model. Our official test scores were 50.41% WER on subtask 1.1, 14.53% WER on subtask 1.3, and 56.15% accuracy on Task 2, ranking third, third, and first, respectively. Across the three systems, the largest gains came from explicit control of alignment, temporal resolution, and evidence diversity rather than from model scale alone.

1 Introduction

NADI 2026 evaluates Arabic speech systems under noise, code switching, and domain mismatch (Sullivan et al., 2026). We participated in three subtasks. Subtask 1.1 requires transcription of noisy speech from eight Arab countries, subtask 1.3 targets Tunisian speech containing Arabic, French, and English, and Task 2 assigns speech to one of 20 country-level dialect or Modern Standard Arabic (MSA) labels. The official ranking metrics are word error rate (WER) for the ASR subtasks and accuracy for dialect identification. The large development–test gaps reported at NADI 2025 illustrate the difficulty of the domain shift (Talafha et al., 2025; Albawwab and Qawasmeh, 2025; Elleuch et al., 2025b).

Our systems share a design principle rather than a common network. For ASR, we combine recognizers from different architecture families by Recognizer Output Voting Error Reduction (ROVER), while controlling the reference hypothesis used for alignment. For dialect identification, we combine complementary acoustic and lexical predictions and then average two independently trained classifiers. This distinction matters: useful diversity came from different error mechanisms, not simply from adding more checkpoints.

Spoken Arabic dialect identification has traditionally used acoustic, phonotactic, and lexical evidence (Ali et al., 2016; Khurana et al., 2017). Recent NADI dialect identifiers have fused ECAPA and WavLM representations or fine-tuned a Whisper encoder with attention pooling (Albawwab and Qawasmeh, 2025; Elleuch et al., 2025b). ASR transcripts have also served as lexical input for Arabic dialect classification (Malmasi and Zampieri, 2016). Our Task 2 system combines these directions by pairing an Arabic self-supervised acoustic branch with two independently generated transcript views. For Tunisian ASR, TEDxTN provides a recent code-switched reference point (Bougares et al., 2025); our analysis isolates the CTC frame-rate and ROVER alignment choices that governed the submitted systems.

Across the ASR systems, the temporal output rate of a CTC encoder became a silent feasibility constraint on fast speech, and the ROVER alignment anchor proved to be a consequential hyperparameter. Shallow language-model fusion helped only when its text added lexical coverage beyond the acoustic model's transcripts. Task 2 instead benefited from dual-ASR lexical evidence, multi-query attentive statistics pooling, and transcript-view dropout. The resulting Task 2 ensemble ranked first; both ASR systems ranked third on tightly contested leaderboards.

2 Tasks and Data

Table 1 summarizes the data used. For subtask 1.1, the provided training set contains 12,800 ut-

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 95–109
October 28-29, 2026 ©2026 Association for Computational Linguistics

Task	Training used	Validation	Metric
1.1	12.8k / 13.6h + 1.3	official	WER ↓
1.3	25k / ∼38h + 4.2k	732 utt.	WER ↓
2	118.8k / 265.9h	10.8k / 25.0h	Acc. ↑

Table 1: Data and primary metric. The 1.1 addition is provided cross-task speech; the 1.3 addition is external labeled speech.

terances (13.6 hours) across Algeria, Egypt, Jordan, Mauritania, Morocco, Palestine, UAE, and Yemen. The final anchor also uses the provided subtask 1.3 training speech. For subtask 1.3, we combine roughly 25,000 provided Tunisian utterances (about 38 hours) with 4,200 labeled utterances from TUNIFRA (Choux et al., 2025) and a Tunisian SLURP-derived corpus (Elleuch et al., 2026a); LinTO Tunisian text (Naouara et al., 2025) is used only for language-model training. Both ASR entries use the open-data, under-5B-parameter condition.

For Task 2 training, we use a public 20-label mixture of 118,762 utterances (265.9 hours), assembled from Micro ADI-20 (Elleuch et al., 2025a) and seven additional Arabic speech resources. The official development split contains 10,806 utterances (25.0 hours). Source-level composition and sampling are reported in Appendix A. ASR text normalization reproduces the organizers' scoring function. Task 2 audio is converted to mono, resampled to 16 kHz, and limited to 30 seconds; transcript normalization applies Unicode NFKC normalization, removes tatweel, and compacts whitespace.

3 Systems

3.1 Subtasks 1.1 and 1.3: heterogeneous ROVER

The ASR pool spans Whisper large-v3 (Radford et al., 2023), MMS-1B (Pratap et al., 2024), the w2v-BERT-2.0 encoder from SeamlessM4T (Barrault et al., 2023), NeMo FastConformer (Rekesh et al., 2023), Cohere Transcribe Arabic (Cassini et al., 2026), and Qwen3-ASR-1.7B (Shi et al., 2026). This deliberately mixes frame-synchronous CTC models with autoregressive and transducer systems. Their normalized hypotheses are combined by equal-weight ROVER (Fiscus, 1997).

The final subtask 1.1 system uses the country partition to route five easier countries to a global three-recognizer committee and Algeria, Mauritania, and Yemen to country-specific committees. Routing requires no dialect identification step: the organizers supply a country field on every development and test utterance, and the submission format is itself one transcript file per country, so the partition is given rather than predicted. Every committee is ordered so its strongest recognizer anchors the ROVER alignment. The final subtask 1.3 system uses a multi-hypothesis committee anchored by a 50 Hz w2v-BERT-2.0 CTC model. A 4-gram KenLM (Heafield, 2011) supplies shallow fusion for CTC members. Appendix B gives the retained members and decoding settings.

For subtask 1.1, we evaluate every committee subset of size two to four on each country's development slice and order the retained members by global development WER. Subtask 1.3 committee changes are checked by paired bootstrap resampling with 2,000 samples (Koehn, 2004). Voting remains equal-weight in both systems.

Appendix B.2 reports the cross-model comparison of parameter scale and encoder adaptation.

3.2 Task 2: acoustic and transcript evidence

The acoustic branch initializes Ara-BEST-RQ-300M (Elleuch et al., 2026b). Its Conformer representations pass through learned attention pooling and a 20-way classifier. After supervised fine-tuning, the branch is frozen and its class logits are cached.

Each utterance is also transcribed independently by frozen Cohere Transcribe Arabic 07-2026 and Whisper large-v3. Within each system member, the Cohere and Whisper transcripts are processed independently by the same MARBERTv2 encoder (Abdul-Mageed et al., 2021) and classification head. The first eight Transformer blocks remain frozen, while the final four (zero-indexed 8–11) are fine-tuned. For each transcript, eight learned queries use unscaled dot-product attention to compute weighted means and standard deviations (Okabe et al., 2018). These statistics pass through layer normalization, a 512-dimensional residual MLP, and a 20-way classifier. A diversity loss discourages different queries from learning identical attention patterns.

At inference, the acoustic, Cohere, and Whisper branches are combined in log-probability space with weights $(0.70, 0.15, 0.15)$. For class k, this is

Task	Final system	Score	Rank
1.1	routed ROVER	50.41 WER	3 / 9
1.3	multi-system ROVER	14.53 WER	3 / 6
2	two-model ensemble	56.15 Acc.	1st

Table 2: Official test results and final ranks.

the normalized weighted geometric pool

$$p_{\text{fuse}}(k \mid x) = \frac{\prod_{m\in\{a,c,w\}} p_m(k \mid x)^{w_m}}{\sum_{j=1}^{20} \prod_{m\in\{a,c,w\}} p_m(j \mid x)^{w_m}},$$
$$(w_a, w_c, w_w) = (0.70, 0.15, 0.15). \tag{1}$$

The final submission averages two independently trained eight-query members. The first is trained to classify the two transcript views separately. The second is trained on the fused prediction with an auxiliary transcript-branch loss and transcript-view dropout (Neverova et al., 2016). During that training, Cohere is removed from the fusion with probability 0.15, Whisper with probability 0.15, and neither with probability 0.70; both are never removed together. At inference all three branches are present, and the members' probability distributions are averaged arithmetically. Figure 1 summarizes the system; Appendix A.2 gives the exact pooling, regularization, and objective definitions.

4 Results and Analysis

4.1 Official results

Table 2 reports the final leaderboard results. Thakaa placed third in both ASR subtasks: the 1.1 score was 0.12 WER behind second place, while the top three 1.3 systems were separated by only 0.12 WER. The Task 2 ensemble achieved 56.15% accuracy and ranked first.

4.2 ASR findings

For a CTC target sequence $y = (y_1, \ldots, y_U)$ and an encoder output of length T, alignment is possible only if

$$T \geq T_{\min} = U + \sum_{u=2}^{U} \mathbf{1}[y_u = y_{u-1}],$$

since consecutive identical target symbols require an intervening blank (Graves et al., 2006). Thus, the relevant constraint is not simply the average number of frames per character, but the temporal margin available relative to $T_{\min}$. On subtask 1.3, the default w2v-BERT adapter reduced the encoder output rate from 50 to 25 Hz, corresponding to approximately 1.8 versus 3.6 encoder frames per reference character on this fast-speech corpus. Although this ratio is not itself a CTC feasibility criterion, it indicates substantially less temporal slack at 25 Hz.

After the 25 Hz run had plateaued at 44.8% validation WER for 16 epochs, we resumed training from that run's encoder weights with the adapter disabled, which doubled the output frame rate, and with a newly initialized CTC head, keeping the training data and schedule unchanged. The resumed run reached 31.5% within 6.6 additional epochs (Figure 2). Continued training and language-model decoding yielded 22.1% WER for the strongest individual system. Together with the head-control experiments in Appendix D.1, these results support temporal resolution as the primary factor responsible for the improvement in this comparison, although the resumed run also received additional encoder updates.

ROVER anchor identity mattered in both tasks. Anchoring a subtask 1.1 committee on its second-strongest member scored 59.0% official test WER, worse than the 51.0% global baseline it was built to improve, against 52.0% correctly ordered; re-anchoring the same 1.3 committee gave 17.11% rather than 17.90% validation WER. Ensembling helped most where the anchor itself was weak: country-specific committees gained on the three hardest 1.1 dialects, whereas on the five easier ones a two-member committee (the anchor plus any single voter) cost 0.5–2 pp and the best three-member committee came within 0.05 pp of the anchor without improving on it, which motivated the final routing policy. KenLM fusion was nearly neutral on 1.1 but reduced one 1.3 CTC system from 54.1% to 42.6% validation WER. The 1.3 LM included text not seen during acoustic fine-tuning, whereas the 1.1 LM largely repeated its transcripts.

4.3 Dialect-identification progression

Table 3 contains systems submitted to the official evaluator. The Cohere transcript configuration scored above its matched Whisper counterpart, and using both transcript views reached 52.96%, compared with 50.11% for the acoustic system. The best single system used four queries, fusion CE, and view dropout, reaching 55.47%. The eight-query branch-CE and fusion-CE-with-dropout systems each scored 55.35%, while their equal probability ensemble reached 56.15%, 0.68 percentage points above the best single system. More queries

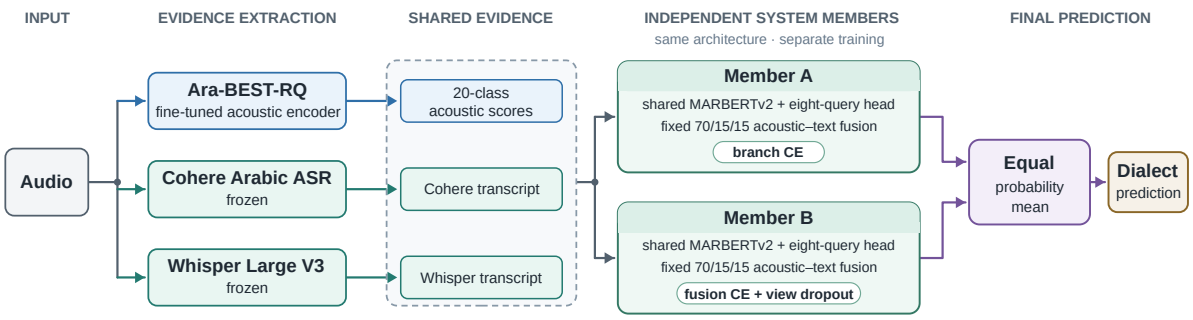


Figure 1: Task 2 final inference pipeline. Acoustic evidence is combined with independently generated Cohere and Whisper transcripts. Two separately trained eight-query members share the architecture but use different objectives; their probability distributions are averaged.

Submitted Task 2 system	Accuracy (%)
Acoustic only	50.11
Acoustic + Whisper	51.14
Acoustic + Cohere	52.05
Acoustic + Cohere + Whisper	52.96
Four-query fusion CE + view dropout	55.47
Eight-query branch CE	55.35
Eight-query fusion CE + view dropout	55.35
Equal ensemble of eight-query systems	**56.15**

Table 3: Selected official Task 2 submissions. The expanded official table, including CostAvg, FPR, and FNR, is in Appendix A.3.

therefore did not consistently improve individual accuracy; the final result came from combining two systems trained with different objectives. The four-query architecture is specified alongside the eight-query version in Appendix A.2.

These are single-run comparisons, so the small differences between individual configurations do not establish a statistically reliable advantage for either query count. Because the acoustic and lexical stages were optimized separately, the first four rows are system comparisons rather than a strict modality-only causal ablation.

4.4 Error patterns and negative results

Code switching was not the bottleneck in the code-switching subtask: on the 1.3 validation set code-switched utterances were easier than Arabic-only ones and Arabic-token error exceeded Latin-token error, so the interventions below addressed what the models already handled best. Frequent residual errors involved Tunisian orthographic variants and extreme speech rates. In subtask 1.1, Maghrebi varieties remained hardest after adaptation. Simulated MUSAN and room-impulse-response augmentation (Snyder et al., 2015; Ko et al., 2017) did not improve 1.1, while speed perturbation, additional out-of-domain acoustic data, pseudo-labeling, and LLM correction did not improve 1.3. Appendix D.5 records these results and their evaluation splits.

4.5 Inference cost of ensembling

Ensembling buys accuracy with compute, so Table 4 reports the real-time factor (RTF, compute time divided by audio duration) of each final ASR system against its strongest single member. Per-member RTFs are measured on one NVIDIA RTX A6000 (48 GB) at batch size 1 with the decoding settings of the submitted runs; the system-level figures are estimated sequential RTFs obtained by summing those component measurements, and for the routed 1.1 system by weighting each member by its utterance coverage under an equal-audio-per-country assumption. Appendix C gives the protocol and per-member breakdown. The two systems occupy different operating points. The subtask 1.1 committee is dialect-routed: Cohere v4 transcribes every utterance as the anchor (RTF 0.058), and each additional voter runs on one to seven of the eight countries, giving an estimated effective RTF of about 0.37, roughly 6.3 times the anchor alone. The global three-member committee costs almost the same (RTF 0.365, 1% less), so the improvement from 51.0% to 50.41% test WER came from routing at essentially unchanged cost; both committees cost about six times Cohere v4 alone. The subtask 1.3 committee runs all ten members on every utterance at RTF 1.24, roughly 83 times its CTC anchor, for a 5.0-point validation gain. In both tasks, the cost is concentrated in the autoregressive voters: the two Whisper members consume 46% of the 1.3 budget while being its two weakest members, and Whisper LoRA accounts for 36% of the 1.1 budget.

Task	System	Members	RTF	×
1.1	routed ROVER (submitted)	7	0.368	6.3
1.1	global ROVER-3	3	0.365	6.3
1.1	Cohere v4 alone	1	0.058	1.0
1.3	ten-system ROVER (submitted)	10	1.244	83
1.3	without both Whisper voters	8	0.667	44
1.3	CTC-only committee	3	0.046	3.0
1.3	CTC-50 Hz alone	1	0.015	1.0

Table 4: Estimated sequential real-time factor (RTF) of the final ASR systems and cheaper alternatives, assembled from per-member measurements on one RTX A6000 at batch size 1 and summed over members run sequentially. The last column is the compute multiplier relative to the anchor recognizer. For 1.1 the RTF is a per-member sum weighted by the share of test utterances each member serves, assuming equal audio per country (Appendix C).

Frame-synchronous CTC members with KenLM fusion are about 19 times cheaper than the Whisper members, and ROVER voting itself is negligible (0.15 ms per utterance). Dropping both Whisper voters approximately halves the 1.3 committee's compute (RTF 1.244 to 0.667, a 46% reduction) and costs about 0.4 validation WER points, and a CTC-only committee (RTF 0.046) still beats every individual system. For Task 2, the summed sequential component RTF is 0.1412 for the best individual system and 0.1433 for the final ensemble, about 1.5% extra computation because the acoustic and ASR outputs are shared (Appendix A.2.5).

5 Conclusion

Our three NADI 2026 systems use structured combinations of diverse evidence. Country-aware ROVER achieved 50.41% WER on noisy Arabic ASR, a frame-rate-aware multi-system ROVER achieved 14.53% on Tunisian code-switched ASR, and an acoustic–dual-transcript ensemble achieved 56.15% accuracy and first place in spoken dialect identification. The experiments highlight temporal feasibility, anchor selection, and cross-representation fusion as practical controls for robust Arabic speech processing.

Limitations

ASR committees cost 6–83 times their anchor recognizer at inference (Section 4.5); the system RTFs are estimates assembled from per-member measurements on one GPU model at batch size 1, so they describe sequential latency rather than peak throughput or a timed end-to-end pipeline. Subtask 1.1 committee selection used the in-domain score of Cohere v4 after the validation split had been folded into that model's training, which may have favored committees containing it. The decision to adopt easy/hard routing was prompted by earlier official test scores (Table 13), although the routing assignment itself was fixed on development data before its single test submission. Subtask 1.3 subset search used one 732-utterance validation split with repeated bootstrap comparisons and no fixed acceptance threshold, so its composition results are exploratory. Each final Task 2 text configuration was trained once, so the comparisons do not quantify variability across training runs. The official evaluator returned aggregate rather than per-dialect errors, which limits conclusions about which dialects benefited from each change. Task 2 inference costs are measured on a small, unbatched sample and summed across sequential components (Appendix A.2.5); they do not establish end-to-end service latency or peak throughput.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105. Association for Computational Linguistics.

Mohammad Abuoudeh. 2026. An acoustical dataset of storytelling and story reading in Jordanian Arabic. *Data in Brief*, 66:112875.

Sanad Albawwab and Omar Qawasmeh. 2025. Lahjati at NADI 2025 a ECAPA-WavLM fusion with multi-stage optimization. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 740–744. Association for Computational Linguistics.

Sadeen Alharbi, Areeb Alowisheq, Zoltán Tüske, Kareem Darwish, Abdullah Alrajeh, Abdulmajeed Alrowithi, Aljawharah Bin Tamran, Asma Ibrahim, Raghad Aloraini, Raneem Alnajim, Ranya Alkahtani, Renad Almuasaad, Sara Alrasheed, Shaykhah Alsubaie, and Yaser Alonaizan. 2024. SADA: Saudi audio dataset for Arabic. In *ICASSP 2024*, pages 10286–10290.

Ahmed Ali, Najim Dehak, Patrick Cardinal, Sameer Khurana, Sree Harsha Yella, James Glass, Peter Bell, and Steve Renals. 2016. Automatic dialect detection in Arabic broadcast speech. In *Interspeech 2016*, pages 2934–2938.

Loïc Barrault, Yu-An Chung, Mariano Coria Meglioli, David Dale, Ning Dong, Paul-Ambroise Duquenne, Hady Elsahar, Hongyu Gong, Kevin Heffernan, John Hoffman, Christopher Klaiber, Pengwei Li, Daniel Licht, Jean Maillard, Alice Rakotoarison, Kaushik Ram Sadagopan, Guillaume Wenzek, Ethan Ye, et al. 2023. SeamlessM4T: Massively multilingual and multimodal machine translation. *arXiv preprint arXiv:2308.11596*.

Fethi Bougares, Salima Mdhaffar, Haroun Elleuch, and Yannick Estève. 2025. TEDxTN: A three-way speech translation corpus for code-switched Tunisian Arabic–English. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 278–287. Association for Computational Linguistics.

Shaun Cassini, Sebastian Vincent, Xiaolu Lu, Julian Mack, Dhruti Joshi, and Pierre Richemond. 2026. Cohere Transcribe Arabic 07-2026. Hugging Face model card.

Alex Choux, Marko Avila, Josep Crego, Fethi Bougares, and Antoine Laurent. 2025. TuniFra: A Tunisian Arabic speech corpus with orthographic transcriptions and French translations. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 64–68. Association for Computational Linguistics.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025a. ADI-20: Arabic dialect identification dataset and models. In *Interspeech 2025*, pages 2775–2779.

Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2026a. SLURP-TN: Resource for Tunisian dialect spoken language understanding. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 1544–1552, Palma de Mallorca, Spain. ELRA Language Resource Association.

Haroun Elleuch, Youssef Saidi, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025b. ELYADATA & LIA at NADI 2025: ASR and ADI subtasks. In *Proceedings of the Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 762–766. Association for Computational Linguistics.

Haroun Elleuch, Ryan Whetten, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2026b. Ara-BEST-RQ: Multi dialectal Arabic SSL. In *ICASSP 2026–2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5326–5330.

Jonathan G. Fiscus. 1997. A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER). In *1997 IEEE Workshop on Automatic Speech Recognition and Understanding Proceedings*, pages 347–354.

Nadia Ghezaiel et al. 2026. SaudiTalk: A multi-source dialectal speech dataset from Saudi Arabia. Hugging Face dataset. Accessed 2026-07-28.

Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. 2006. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376.

Kenneth Heafield. 2011. KenLM: Faster and smaller language model queries. In *Proceedings of the Sixth Workshop on Statistical Machine Translation*, pages 187–197. Association for Computational Linguistics.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Sameer Khurana, Maryam Najafian, Ahmed Ali, Tuka Al Hanai, Yonatan Belinkov, and James Glass. 2017. QMDIS: QCRI-MIT advanced dialect identification system. In *Interspeech 2017*, pages 2591–2595.

Reinhard Kneser and Hermann Ney. 1995. Improved backing-off for M-gram language modeling. In *1995 International Conference on Acoustics, Speech, and Signal Processing*, volume 1, pages 181–184.

Tom Ko, Vijayaditya Peddinti, Daniel Povey, Michael L. Seltzer, and Sanjeev Khudanpur. 2017. A study on data augmentation of reverberant speech for robust speech recognition. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 5220–5224.

Philipp Koehn. 2004. Statistical significance tests for machine translation evaluation. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 388–395. Association for Computational Linguistics.

Shervin Malmasi and Marcos Zampieri. 2016. Arabic dialect identification in speech transcripts. In *Proceedings of the Third Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial3)*, pages 106–113. The COLING 2016 Organizing Committee.

Omer Nacar, Serry Sibaee, Adel Ammar, Yasser Alhabashi, Nadia Samer Sibai, Yara Farouk Ahmed, Ahmed Saud Alqusaiyer, Sulieman Mahmoud AlMahmoud, Abdulrhman Mamdoh Mukhaniq, Lubaba Raed, Sulaiman Mohammed Alatwah, Waad Nasser Alqahtani, Yousif Abdulmajeed Alnasser, Mohamed Aziz Khadraoui, and Wadii Boulila. 2026. ARCADE: A city-scale corpus for fine-grained Arabic dialect tagging. *arXiv preprint arXiv:2601.02209*.

Hedi Naouara, Jean-Pierre Lorré, and Jérôme Louradour. 2025. LinTO audio and textual datasets to train and evaluate automatic speech recognition in Tunisian Arabic dialect. *arXiv preprint arXiv:2504.02604*.

Natalia Neverova, Christian Wolf, Graham W. Taylor, and Florian Nebout. 2016. ModDrop: Adaptive multi-modal gesture recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38(8):1692–1706.

oddadmix. 2026. Lahgtna: Dialect-balanced Arabic ASR (v3 small). Hugging Face dataset. Accessed 2026-07-21.

Koji Okabe, Takafumi Koshinaka, and Koichi Shinoda. 2018. Attentive statistics pooling for deep speaker embedding. In *Interspeech 2018*, pages 2252–2256.

Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, Alexei Baevski, Yossi Adi, Xiaohui Zhang, Wei-Ning Hsu, Alexis Conneau, and Michael Auli. 2024. Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research*, 25(97):1–52.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28492–28518. PMLR.

Dima Rekesh, Nithin Rao Koluguri, Samuel Kriman, Somshubra Majumdar, Vahid Noroozi, He Huang, Oleksii Hrinchuk, Krishna Puvvada, Ankur Kumar, Jagadeesh Balam, and Boris Ginsburg. 2023. Fast conformer with linearly scalable attention for efficient speech recognition. In *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1–8.

Xian Shi, Xiong Wang, Zhifang Guo, Yongqi Wang, Pei Zhang, Xinyu Zhang, Zishan Guo, Hongkun Hao, Yu Xi, Baosong Yang, Jin Xu, Jingren Zhou, and Junyang Lin. 2026. Qwen3-ASR technical report. *arXiv preprint arXiv:2601.21337*.

David Snyder, Guoguo Chen, and Daniel Povey. 2015. MUSAN: A music, speech, and noise corpus. *arXiv preprint arXiv:1510.08484*.

Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, Abdel-Rahim Elmadany, Youssef Mohamed, Salima Mdhaffar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. 2026. NADI-2026: The second multidialectal Arabic speech processing shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Bashar Talafha, Karima Kadaoui, Samar Mohamed Magdy, Mariem Habiboullah, Chafei Mohamed Chafei, Ahmed Oumar El-Shangiti, Hiba Zayed, Mohamedou Cheikh Tourad, Rahaf Alhamouri, Rwaa Assi, Aisha Alraeesi, Hour Mohamed, Fakhraddin Alwajih, Abdelrahman Mohamed, Abdellah El Mekki, El Moatez Billah Nagoudi, Benelhadj Djelloul Mama Saadia, Hamzah A. Alsayadi, Walid Al-Dhabyani, and 8 others. 2024. Casablanca: Data and models for multidialectal Arabic speech recognition. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21745–21758. Association for Computational Linguistics.

Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. 2025. NADI 2025: The first multidialectal Arabic speech processing shared task. In *Proceedings of the Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733.

A Task 2 Data and Reproduction Details

A.1 Public speech mixture

The Task 2 corpus combines the complete Micro ADI-20 training split; selected official-training material from SADA (Alharbi et al., 2024) and Lahgtna-v3-small (oddadmix, 2026); selected SLURP-TN training material (Elleuch et al., 2026a); Casablanca's labeled validation material (Talafha et al., 2024); and qualified ARCADE (Nacar et al., 2026), Layla Witheeb (Abuoudeh, 2026), and SaudiTalk (Ghezaiel et al., 2026) audio. Country or dialect metadata supplied by each source is mapped to the task inventory. These resources span broadcast, radio, read, spontaneous, command, and social-media speech. Table 5 reports the per-source utterance and hour counts.

Source	Utterances	Hours
Micro ADI-20	67,015	163.8
Lahgtna-v3-small	20,604	55.2
SADA subset	16,499	21.7
Casablanca	6,676	8.0
SLURP-TN	2,676	2.8
ARCADE	599	1.6
Layla Witheeb	2,479	6.7
SaudiTalk	2,214	6.0
Total	**118,762**	**265.9**

Table 5: Task 2 system-development corpus. Source hours are rounded.

For Lahgtna-v3-small we retain the official training split for 12 unambiguous country labels and exclude its ambiguous Saudi/Gulf label. The SADA subset retains valid MSA, Egyptian, and Yemeni material and caps Saudi at 10,000 segments while preserving every Shamali and Janubi segment and sampling Najdi and Hijazi. High-confidence

single-speaker ARCADE training recordings, eligible long Casablanca recordings, and the unsplit Layla Witheeb and SaudiTalk recordings are converted to non-overlapping 1–12-second segments. Casablanca's upstream split is named validation but is used as training material here. Task-level class counts range from 2,847 (KUW) to 15,865 (KSA), and duration ranges from 7.1 to 27.4 hours. The official development split contains 10,806 utterances (25.0 hours), with class counts ranging from 197 to 1,649.

A.2 Model and optimization details

The acoustic branch initializes the 300M-parameter Ara-BEST-RQ model. Its convolutional front end and 24-layer Conformer feed learned attention pooling and a bias-free 20-way classifier. The full encoder and classifier are trained with cross-entropy. Acoustic training uses a class-balanced sampler with source cycling and two Adam optimizers, encoder/head learning rates $10^{-5}/10^{-4}$, head weight decay 2×10^{-6}, gradient clipping at 1, bfloat16, and a global batch of 64. The acoustic branch is then frozen and its logits are cached before lexical training.

The ASR models are Cohere Transcribe Arabic 07-2026 and Whisper large-v3. Both are frozen and used only to generate transcripts. Every text variant starts independently from the same pretrained MARBERTv2 model rather than a previously fine-tuned task checkpoint. Its top four blocks and the head are trainable. Tokenization uses dynamic padding, truncation to length 128, and ignores padding tokens during pooling.

A.2.1 Multi-query attentive-statistics head

For transcript view $v \in \{c, w\}$ and the batch-padded token length $T \leq 128$, let

$$H_i^{(v)} = \left(h_{i,1}^{(v)}, \ldots, h_{i,T}^{(v)}\right), \qquad h_{i,t}^{(v)} \in \mathbb{R}^D,$$

denote the MARBERTv2 token representations for utterance i. Each learned query $u_q \in \mathbb{R}^D$, $q = 1, \ldots, Q$, assigns the unscaled dot-product score

$$e_{i,q,t}^{(v)} = u_q^{\mathsf{T}} h_{i,t}^{(v)}. \tag{2}$$

No $1/\sqrt{D}$ scaling is applied. Let $M_{i,t}^{(v)} \in \{0, 1\}$ indicate whether position t is a valid, non-padding token. Attention is normalized over these valid positions:

$$\alpha_{i,q,t}^{(v)} = \frac{M_{i,t}^{(v)} \exp(e_{i,q,t}^{(v)})}{\sum_{s=1}^{T} M_{i,s}^{(v)} \exp(e_{i,q,s}^{(v)})}. \tag{3}$$

For each query, attentive-statistics pooling computes

$$\mu_{i,q}^{(v)} = \sum_{t=1}^{T} \alpha_{i,q,t}^{(v)} h_{i,t}^{(v)} \tag{4}$$

and

$$\nu_{i,q}^{(v)} = \sum_{t=1}^{T} \alpha_{i,q,t}^{(v)} \left(h_{i,t}^{(v)} - \mu_{i,q}^{(v)}\right)^{\odot 2}, \tag{5}$$

$$\sigma_{i,q}^{(v)} = \left[\max\left(\nu_{i,q}^{(v)}, 10^{-8}\right)\right]^{\odot 1/2}. \tag{6}$$

Here the maximum, square, and square root are applied elementwise. The Q mean–standard-deviation pairs are concatenated:

$$r_i^{(v)} = \underset{q=1}{\overset{Q}{\operatorname{concat}}} \left[\mu_{i,q}^{(v)}; \sigma_{i,q}^{(v)}\right] \in \mathbb{R}^{2QD}. \tag{7}$$

Both the four-query and eight-query systems use this definition, with $Q = 4$ and $Q = 8$, respectively. For $D = 768$, the pooled vector has 6,144 dimensions when $Q = 4$ and 12,288 when $Q = 8$. Both use the same 512-dimensional residual projection and 20-class output design below; changing Q changes the query matrix and the input dimensions of the normalization and projection layers. Let $\widetilde{r}_i^{(v)} = \mathrm{LayerNorm}(r_i^{(v)})$. The 512-dimensional text representation and its class logits are

$$\begin{aligned} g_i^{(v)} = {} & \mathrm{Dropout}\Big(\mathrm{GELU}\Big(W_p \widetilde{r}_i^{(v)} + b_p\Big)\Big) \\ & + W_r \widetilde{r}_i^{(v)}, \end{aligned} \tag{8}$$

$$z_i^{(v)} = W_c g_i^{(v)} + b_c. \tag{9}$$

The MARBERTv2 encoder, learned queries, pooling head, and classifier parameters are shared between $v = c$ and $v = w$ within a system member.

A.2.2 Query-diversity regularization

The diversity term regularizes the attention distributions rather than the query parameter vectors. Let $n = 1, \ldots, N$ index the transcript examples in a training batch and define

$$\bar{\alpha}_{n,q} = \frac{\alpha_{n,q}}{\|\alpha_{n,q}\|_2}. \tag{10}$$

The loss is the mean squared cosine similarity over all ordered pairs of distinct queries:

$$\mathcal{L}_{\mathrm{div}} = \frac{1}{NQ(Q-1)} \sum_{n=1}^{N} \sum_{\substack{q,r=1 \\ q \neq r}}^{Q} \left(\bar{\alpha}_{n,q}^{\mathsf{T}} \bar{\alpha}_{n,r}\right)^2. \tag{11}$$

Each utterance contributes both transcript views, so $N = 2B$ for an utterance batch of size B. Both final members use diversity coefficient $\lambda_{\text{div}} = 0.01$.

Text training traverses all 118,762 utterances and creates one Cohere and one Whisper view per item. AdamW uses encoder/head learning rates $10^{-5}/10^{-4}$, weight decay $10^{-2}/10^{-5}$, gradient clipping at 1, bfloat16, and global utterance batch 64. Dropout in the nonlinear projection is 0.2. The four-query head contains 6.3M parameters and the eight-query head contains 12.6M parameters.

A.2.3 Fusion and training objectives

Let $p_j(k \mid x_i)$ be the softmax-normalized class probability from branch $j \in \{a, c, w\}$, corresponding to acoustic, Cohere, and Whisper evidence, and let $(w_a, w_c, w_w) = (0.70, 0.15, 0.15)$. The base dual-ASR fusion score and distribution are

$$s_{i,k} = \sum_{j \in \{a,c,w\}} w_j \log p_j(k \mid x_i), \quad (12)$$

$$p_{\text{fuse}}(k \mid x_i) = \frac{\exp(s_{i,k})}{\sum_{r=1}^{20} \exp(s_{i,r})}. \quad (13)$$

This is the normalized geometric pool in Eq. 1. Submitted single-ASR systems instead combine acoustic and text log probabilities with weights 0.75 and 0.25.

Let $y_i \in \{1, \ldots, 20\}$ be the target class. The branch-classification loss is

$$\mathcal{L}_{\text{branchCE}} = -\frac{1}{2B} \sum_{i=1}^{B} \sum_{v \in \{c,w\}} \log p_v(y_i \mid x_i). \quad (14)$$

The first final member minimizes

$$\mathcal{L}_A = \mathcal{L}_{\text{branchCE}} + 0.01 \mathcal{L}_{\text{div}}. \quad (15)$$

This is equivalent to one mean cross-entropy after interleaving the B Cohere and B Whisper examples into a batch of size $2B$. The frozen acoustic branch does not contribute to $\mathcal{L}_{\text{branchCE}}$.

Transcript-view dropout is applied only to the fusion term of the second member. For each utterance, let $D_i \in \{c, w, \varnothing\}$ identify the transcript view removed from the fusion. We sample

$$\begin{aligned} \Pr(D_i = c) &= \Pr(D_i = w) = 0.15, \\ \Pr(D_i = \varnothing) &= 0.70. \end{aligned} \quad (16)$$

Thus one transcript view is removed with total probability 0.30, and both are never removed together. Define retention indicators $m_{i,a} = 1$, $m_{i,c} = \mathbf{1}[D_i \neq c]$, and $m_{i,w} = \mathbf{1}[D_i \neq w]$. The renormalized weights are

$$\widetilde{w}_{i,j} = \frac{m_{i,j} w_j}{\sum_{\ell \in \{a,c,w\}} m_{i,\ell} w_\ell}. \quad (17)$$

The dropout-conditioned score and distribution are

$$s_{i,k}^{(D_i)} = \sum_{j \in \{a,c,w\}} \widetilde{w}_{i,j} \log p_j(k \mid x_i), \quad (18)$$

$$p_{\text{fuse}}^{(D_i)}(k \mid x_i) = \frac{\exp(s_{i,k}^{(D_i)})}{\sum_{r=1}^{20} \exp(s_{i,r}^{(D_i)})}. \quad (19)$$

The second member's primary and auxiliary losses are

$$\mathcal{L}_{\text{fuseCE}} = -\frac{1}{B} \sum_{i=1}^{B} \log p_{\text{fuse}}^{(D_i)}(y_i \mid x_i), \quad (20)$$

$$\mathcal{L}_{\text{aux}} = \mathcal{L}_{\text{branchCE}}, \quad (21)$$

and its complete objective is

$$\mathcal{L}_B = \mathcal{L}_{\text{fuseCE}} + 0.25 \mathcal{L}_{\text{aux}} + 0.01 \mathcal{L}_{\text{div}}. \quad (22)$$

Both transcript views remain present in $\mathcal{L}_{\text{aux}}$ and $\mathcal{L}_{\text{div}}$. At inference, no view is dropped and the base $(0.70, 0.15, 0.15)$ weights are restored.

A.2.4 Final two-model ensemble

Let $s_A(x), s_B(x) \in \mathbb{R}^{20}$ denote the fused class-score vectors saved by the independently trained branch-CE and fusion-CE members. Their member-level probabilities are

$$\begin{aligned} p_A(\cdot \mid x) &= \text{softmax}(s_A(x)), \\ p_B(\cdot \mid x) &= \text{softmax}(s_B(x)). \end{aligned} \quad (23)$$

The final ensemble uses the arithmetic mean

$$p_{\text{ens}}(k \mid x) = \tfrac{1}{2} p_A(k \mid x) + \tfrac{1}{2} p_B(k \mid x), \quad (24)$$

and predicts

$$\widehat{y} = \operatorname*{argmax}_{k \in \{1,\ldots,20\}} p_{\text{ens}}(k \mid x). \quad (25)$$

No parameters are learned for this final ensemble. This arithmetic averaging is distinct from the normalized geometric branch fusion within each member.

A.2.5 Inference computation and reuse

The final Task 2 ensemble reuses one acoustic prediction and the two ASR transcripts for both members. It therefore requires one Ara-BEST-RQ pass, one Cohere transcription, and one Whisper transcription per utterance. Each member has its own partially fine-tuned MARBERTv2 parameters and evaluates both transcripts, giving four transcript encodings for the ensemble versus two for a single dual-ASR member. The acoustic and ASR computations are shared; the added work is the second text classifier applied to both views and a 20-class probability average. The four-query single system also has a smaller head than either eight-query ensemble member.

We measured these components separately on 80 uniformly sampled released test utterances, selected without labels using seed 20260910. The sample contains 578.79 s of original audio; two clips exceed 30 s, leaving 530.58 s after the existing center-crop policy. Each model uses three disjoint utterances for initial priming, followed by three complete warm-up passes over the 80-utterance workload before ten measured repeats. We use batch size one on one RTX A6000 with four CPU threads. The environment is PyTorch 2.9.1, CUDA 12.8, and Transformers 5.4.0. The acoustic and Whisper paths use float16 inference, and Cohere and the text path use bfloat16; all weights and inference settings are frozen. Both ASR decoders use their saved generation configuration with Arabic output and a 128-token cap; MARBERTv2 truncates at 128 tokens.

Timers synchronize the GPU and include feature extraction, transfers, ASR generation, text normalization and tokenization, classification, and fusion. Model loading, file I/O, warm-up, and artifact writing are excluded. Waveform decoding, resampling, and cropping were measured separately and add 0.00003 RTF when counted once. Table 6 sums the required component times and divides by the processed audio duration, counting the shared front end only once. These are reconstructed sequential-computation costs, not directly timed end-to-end service latencies or measurements of batched throughput. The Task 1 timings in Section 4.5 are not reused for Task 2.

After warm-up, acoustic passes range from 2.43 to 2.55 s, with a mean of 2.50 s. The final ensemble costs about 1.5% more than the best single system in this protocol. This small increment depends on sharing the acoustic and ASR outputs; it is not a general latency claim across hardware, batching, or implementations.

Task 2 system	Mean RTF	Repeat range
Acoustic branch alone	0.0047	0.0046–0.0048
Best single ($Q = 4$)	0.1412	0.1405–0.1433
Branch CE ($Q = 8$)	0.1411	0.1404–0.1433
Fusion CE + dropout ($Q = 8$)	0.1412	0.1406–0.1433
Final two-member ensemble	0.1433	0.1427–0.1455

Table 6: Task 2 inference cost: sum of sequential component wall times divided by 530.58 s of processed audio, over ten measured repeats after full-workload warm-up.

A.3 Expanded official Task 2 results

Table 7 reports the systems analyzed in this paper, with the accuracy, cost average, false-positive rate, and false-negative rate returned by the official evaluator.

B ASR System Inventory and Decoding

B.1 Individual recognizers

Table 8 gives the strongest members of each pool.

The final 1.1 selection pool contains seven systems: Cohere Arabic v4, three checkpoint or decoding variants of NeMo FastConformer, NeMo CTC with KenLM, Whisper large-v3 with rank-64 LoRA on the attention projections and output projection (Hu et al., 2022), and Qwen3-ASR-1.7B. The final 1.3 committee contains ten systems anchored on the strongest 50 Hz w2v-BERT CTC model. Three are frame-synchronous CTC voters decoded with LM shallow fusion: two checkpoints from the 50 Hz w2v-BERT lineage and one MMS-1B system. A third checkpoint from that same CTC lineage was tested and added nothing, indicating that the lineage had saturated. The remaining seven are autoregressive hypotheses from the Cohere, Qwen3-ASR, and Whisper fine-tunes: three Cohere checkpoints (stage-2, a 1,600-step checkpoint, and an annealed variant), two Qwen3-ASR checkpoints, and prompted and unprompted decodes of one Whisper fine-tune. We spanned architecture families rather than stacking checkpoints of one model because every autoregressive decoder we trained eventually showed repetition or paraphrase-loop degeneration, a failure mode that frame-synchronous CTC models avoid because they have no autoregressive feedback loop; they can still emit wrong or duplicated words, but not runaway loops. The largest single system in either

Submitted system	Accuracy (%)	Cost	FPR	FNR
Acoustic only	50.11	0.1291	**0.0215**	0.1076
Acoustic + Whisper	51.14	0.1192	0.0255	0.0937
Acoustic + Cohere	52.05	0.1139	0.0250	0.0889
Acoustic + Cohere + Whisper	52.96	0.1142	0.0263	0.0879
Four-query branch CE	54.90	0.1128	0.0249	0.0879
Four-query fusion CE	54.78	0.1093	0.0252	0.0842
Four-query fusion CE + view dropout	55.47	0.1094	0.0252	0.0842
Eight-query branch CE	55.35	0.1132	0.0246	0.0886
Eight-query fusion CE + view dropout	55.35	**0.1088**	0.0253	**0.0835**
Equal ensemble of final two systems	**56.15**	0.1131	0.0228	0.0903

Table 7: Official Task 2 test results returned by CodaBench. Every row is an uploaded submission with a recorded evaluator return.

System	Type	Dev WER	Task
CTC-50 Hz (w2v-BERT)	CTC	22.1	1.3
Cohere stage 2	enc–dec	22.7	1.3
Qwen3-ASR-1.7B	audio-LLM	28.0	1.3
MMS-1B + CTC	CTC	29.6	1.3
Cohere Arabic v4[†]	enc–dec	38.3[†]	1.1
NeMo FastConf. v7-avg	RNNT/CTC	55.1	1.1
NeMo FastConf. v2 + LM	CTC	56.6	1.1
Whisper large-v3 LoRA	enc–dec	59.4	1.1

Table 8: Strongest ensemble members on the official development splits. [†]The 1.1 validation split was folded into this model's training, so this row is in-domain and is not directly comparable to the others.

pool is 2B parameters, placing both entries in the under-5B track.

For subtask 1.3, the final system uses neither LinTO acoustic data nor pseudo-labeled speech. We also declined to use the public source videos and captions from which the evaluation material was derived. Text normalization matches the organizers' scoring function character-for-character; this check identified and corrected an earlier pipeline discrepancy that removed the apostrophe in French contractions.

The settings below are those of the subtask 1.3 submission; the subtask 1.1 submission used cheaper per-member settings, listed in Appendix C. Autoregressive systems use beam search with 10 beams (five for selected Cohere checkpoints), length penalty 0.9, repetition penalty 1.03, and no-repeat 3-grams; the strongest Cohere stage-2 decode uses 0.8 and 1.05 for the first two penalties. CTC systems use `pyctcdecode` with a 4-gram KenLM at LM weight 0.5 and word insertion bonus 1.5. The subtask 1.1 NeMo CTC and RNNT decoders use beam widths 128 and 10, respectively. The unigram list is passed explicitly because `pyctcdecode` cannot recover it from a binary KenLM.

B.2 Adaptation versus model scale

In the comparisons we ran, parameter count alone predicted adapted performance poorly, whereas adapting the acoustic encoder mattered in each case; we report these as empirical observations rather than a general claim. On subtask 1.1, a 2B zero-shot model reached 79% WER while a fully fine-tuned 120M FastConformer reached 55%. On subtask 1.3, a 7.8B model with only its 1.3B-parameter decoder fine-tuned reached 41.0%, compared with 28.0% for a fully adapted 1.7B model.

B.3 ROVER committees

For subtask 1.1, all subsets of size 2–4 were searched on each country's development slice and ordered by the strongest global-development recognizer. Table 9 gives the resulting committees.

Country	Final committee
Egypt, Jordan, Morocco, Palestine, UAE	Cohere v4 + NeMo v7-avg + Whisper LoRA
Algeria	Cohere v4 + NeMo v2-LM + Qwen3-ASR
Mauritania	Cohere v4 + NeMo v5-avg + Whisper LoRA + Qwen3-ASR
Yemen	Cohere v4 + NeMo v5 + Whisper LoRA

Table 9: Final subtask 1.1 committees.

C Inference Cost Measurements

RTF was measured on one NVIDIA RTX A6000 (48 GB) in the project's PyTorch environment, using a fixed random sample of 80 utterances per task (316 s of subtask 1.1 validation audio; 258 s of subtask 1.3 test audio, whose mean duration of 3.23 s

matches the 3.08 s of the full test set). Every member ran in a separate process with a three-utterance warm-up, CUDA synchronization around the timed region, batch size 1, and the decoding settings used to generate the submitted hypotheses. For subtask 1.3 these are the settings of Appendix B. The subtask 1.1 submission used different, cheaper settings, listed per member in Table 10: Cohere v4 at beam 4, the NeMo RNNT and CTC decoders at beam widths 10 and 128, and greedy decoding for Whisper LoRA and Qwen3-ASR. Timing includes feature extraction, the model forward pass, and beam search or LM-fused decoding; audio loading and model loading are excluded. Loading all members takes 59 s for 1.1 and 108 s for 1.3, amortized over the test set. Each configuration was timed once; the three NeMo RNNT checkpoints, which share an architecture and decoder, measured 0.149, 0.149, and 0.156, which bounds checkpoint-to-checkpoint variation at about 5% but is not a repeated-run uncertainty estimate. No quantization, compilation, or batching was applied, so the autoregressive members in particular could be made cheaper. All figures are specific to this GPU, batch size, and decoding configuration.

Table 10 breaks down the routed subtask 1.1 system. Because each voter serves a subset of countries, its effective contribution is approximated as its RTF multiplied by the fraction of the balanced 4,000-utterance test set it covers; this assumes equal audio duration per country, and the total is a sum of per-member estimates rather than a directly timed routed pipeline. Cohere v4 is both the strongest member and, at beam 4, cheaper than every autoregressive voter; Whisper LoRA is the dominant cost because it serves seven of eight countries. Extrapolating from the sample, transcribing the 3.7-hour test set is estimated at approximately 81 minutes of GPU time for the committee against 13 minutes for Cohere v4 alone. On unrounded measurements the routed system costs 1.0% more than the global three-member committee (0.3681 against 0.3645), so the 0.6-point test gain of routing came from reassigning voters rather than from adding compute.

Table 11 gives the subtask 1.3 breakdown by configuration. We assume that members sharing an architecture and decoding settings have the same cost and timed one checkpoint per configuration. This is an approximation: autoregressive checkpoints can produce different output lengths and therefore different runtimes, and the 1.1 NeMo checkpoints above varied by about 5%, so the 1.3 totals carry at least that uncertainty. The CTC members are non-autoregressive, and their KenLM beam-100 decode adds about 0.003 RTF per CTC member, 0.24% of the ensemble each and under 1% for all three, for the 20% relative WER gain reported in Appendix D.4. Table 12 re-scores nested committees on the validation split with the same CTC-50 Hz anchor and equal weights as the submission. Only the submitted ten-member configuration was evaluated on the official test set (14.53%); no test scores are reported for the other rows.

Member	Decode	Cov.	RTF	Eff.
Cohere v4 (anchor)	beam 4	1.000	0.058	0.058
NeMo v2 + LM	CTC beam 128	0.125	0.026	0.003
NeMo v5	RNNT beam 10	0.125	0.149	0.019
NeMo v5-avg	RNNT beam 10	0.125	0.149	0.019
NeMo v7-avg	RNNT beam 10	0.625	0.156	0.097
Whisper LoRA	greedy	0.875	0.150	0.132
Qwen3-ASR	greedy	0.250	0.162	0.040
ROVER (CPU)	–	1.000	<0.001	<0.001
Effective total				**0.368**

Table 10: Subtask 1.1 per-member RTF with the submitted decoding settings. Cov. is the fraction of test utterances the member serves under the routing of Table 9; Eff. is RTF × Cov.

Configuration	Params	n	RTF	Share
w2v-BERT CTC-50 Hz + LM	606M	2	0.015	2.4%
MMS-1B CTC + LM	965M	1	0.016	1.3%
Cohere beam 5	2.07B	3	0.083	19.9%
Qwen3-ASR beam 10	2.04B	2	0.187	30.0%
Whisper-v3 beam 10	1.54B	2	0.288	46.4%
ROVER (CPU)	–	–	<0.001	<0.1%
Sequential total		**10**		**1.244**

Table 11: Subtask 1.3 RTF by configuration, one checkpoint timed per configuration; n is the number of committee members with that configuration. With one RTX A6000 per member, the idealized parallel bound is the Whisper RTF of 0.29; this is a compute bound, not a measured deployment.

D ASR Diagnostic Experiments

D.1 CTC temporal feasibility

Let $y = (y_1, \ldots, y_U)$ denote a CTC target sequence of length U and let T denote the number of encoder output frames. Under the standard CTC

Committee	Dev WER	RTF
10 members (submitted, re-scored)	17.09	1.244
9: drop unprompted Whisper	17.25	0.955
8: drop both Whisper	17.48	0.667
6: drop Whisper and Qwen3	18.36	0.294
3: CTC members only	21.64	0.046
1: CTC-50 Hz anchor alone	22.13	0.015

Table 12: Subtask 1.3 validation WER against compute for nested committees, all anchored on the CTC-50 Hz model with equal weights. The submitted committee's recorded validation score is 17.11; the 17.09 here is the same ten members re-scored by the ablation script, and the 0.02 difference comes from the order in which the non-anchor hypotheses are passed to ROVER, which we did not control in the original run.

collapse operator, a valid alignment exists only if

$$T \geq T_{\min} = U + R(y),$$
$$R(y) = \sum_{u=2}^{U} \mathbf{1}[y_u = y_{u-1}].$$

where $R(y)$ counts adjacent repeated target symbols. Each target symbol requires at least one frame, and each adjacent repetition requires at least one additional separating frame.

Subtask 1.3 contains approximately 14 reference characters per second on average, so the 25 Hz and 50 Hz encoder output rates correspond to the ratios of approximately 1.8 and 3.6 frames per reference character discussed in Section 4.2; utterances that are faster or denser than average approach or violate their individual $T_{\min}$. Figure 2 shows the controlled intervention. The recipe this derives from (Bougares et al., 2025) changes the frame rate *and* inserts transformer layers before the CTC head; we separated the two, and the head change alone was neutral or harmful across three evaluations, leaving temporal resolution as the operative change. The failure is also silent: no error and no warning, only a plateau that reads as convergence. That a filter discarding targets which cannot fit the available frames has to exist at all is corroborating evidence; ours was miscalibrated by exactly $2\times$, so infeasible rows passed through, and because the CTC loss was configured with `zero_infinity` enabled, their infinite losses were zeroed and contributed no gradient instead of raising an error. A more direct pre-training diagnostic is therefore to compute $T/T_{\min}$ for every training example rather than relying only on an average frames-per-character ratio. Examples with $T < T_{\min}$ are strictly infeasible under CTC, while values only modestly above one indicate little alignment slack. In our experiments, increasing the average ratio from approximately 1.8 to 3.6 frames per character was associated with substantially better optimization on this fast-speech corpus. We therefore treat a low frames-per-character ratio as an empirical warning of limited temporal resolution, not as a universal CTC feasibility threshold. For context, TEDxTN (Bougares et al., 2025) reported 19.9% WER on its validation segments; our strongest individual CTC system reached 22.1% and our ensemble 17.1%, but we have not confirmed that the two evaluations use the same utterances and text normalization, so this is a contextual rather than a controlled comparison.

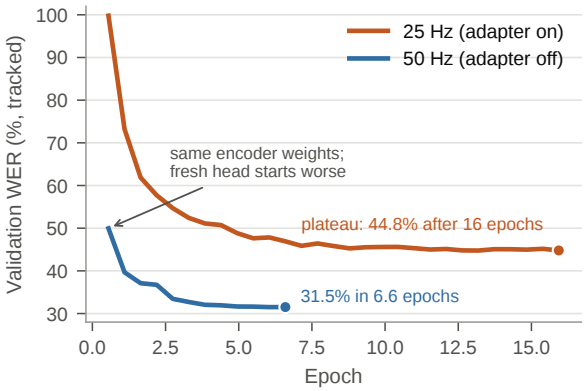


Figure 2: Subtask 1.3 validation trajectories. The 25 Hz run trains from the pretrained encoder with the adapter enabled. The 50 Hz run resumes from the encoder weights of that run at its 16-epoch plateau, with the adapter disabled and a newly initialized CTC head, on the same data and schedule; its epoch axis counts epochs after the restart. The fresh head starts above the 44.8% plateau it resumes from and crosses below it within two epochs.

D.2 Anchor identity and ensemble regime

Table 13 collects the anchor and routing experiments, with the evaluation split marked on each row. For subtask 1.1, the decision to route easy and hard countries differently was prompted by the official test scores of the earlier rows, but the routing itself, which countries share the global committee and which committee each hard country receives, was chosen on development data alone and then submitted once.

Figure 3 plots, for each country, the development WER of the final routed committee of Table 9 minus that of the Cohere v4 anchor alone. Two comparisons must be kept apart. The figure shows the *selected* committees: on the three hardest

Configuration	WER (%)
1.1 global ROVER-3	51.0 test
1.1 dialect-aware, NeMo anchor	59.0 test
1.1 same committees, Cohere anchor	52.0 test
1.1 + easy/hard routing	**50.41 test**
1.3 committee, Cohere anchor	17.90 dev
1.3 same committee, CTC-50 Hz anchor	**17.11 dev**

Table 13: Anchor and routing experiments. Evaluation splits are shown explicitly; intermediate 1.1 submissions were displayed to two decimals.

countries the country-specific committees gained 1.7–5.25 pp, while on the five easiest the selected three-member global committee lies within 0.05 pp of the anchor, so those points sit on the zero line. The forced two-member committees that the search also evaluated, the anchor plus any single voter, cost 0.5–2 pp on the same five countries and are not plotted. The crossover near 40% anchor WER is descriptive rather than universal, but it explains why additional voters helped difficult dialects and could at best be made neutral on easy ones. Because Cohere v4 trained on the 1.1 development split, the anchor WERs on the horizontal axis are in-domain, and the plotted differences are the values the committee search saw rather than unbiased estimates. On subtask 1.3, committee changes were checked by paired bootstrap resampling with 2,000 samples (Koehn, 2004); we did not fix a decision threshold or correct for repeated comparisons on the one validation split, so the 1.3 committee-search results are exploratory.

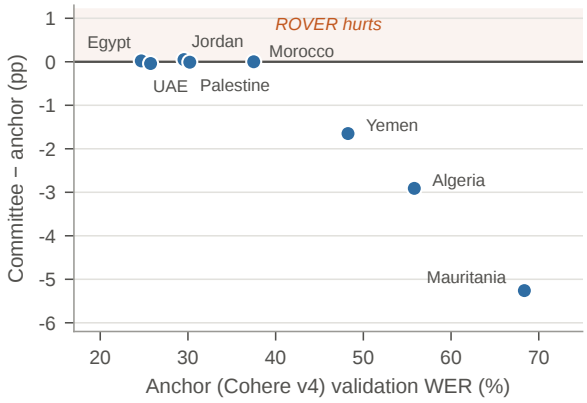


Figure 3: Subtask 1.1 development data: WER of the final routed committee minus WER of the Cohere v4 anchor alone, against the anchor's country-specific development WER. Cohere v4 trained on this development split, so the anchor WERs are in-domain and the differences are those seen during committee selection.

A complementary experiment combined four successive continual fine-tunes from one FastConformer lineage. The best committee exactly tied the strongest individual system and produced no net-positive corrections, indicating that ROVER consumed architectural error diversity rather than voter count alone.

D.3 Weighting, metrics, and transfer

Weighted voting did not improve either subtask. These experiments predate the seven-system selection pool of Appendix B and run over all 20 hypotheses we had generated at that point, including zero-shot systems as weak as 78.9% WER. On subtask 1.1 development data, that 20-hypothesis committee scored 52.20% WER with uniform weights, compared with 52.52% under inverse-WER weighting and 54.76% under country-aware top-5 selection. On subtask 1.3, inverse-WER weighting was effectively unchanged from uniform voting.

ROVER affected the two error metrics differently. From the strongest individual subtask 1.3 system to the committee, WER fell by approximately 23% relative but character error rate (CER) by only 6.8%; the CER/WER ratio increased from 0.368 to 0.443. Across five paired validation/test measurements, gains from improved component models transferred at approximately 1.0–1.2:1, whereas committee-composition changes transferred at roughly 0.62–0.65:1; that is, about 35–38% of a validation gain obtained by exhaustive subset selection on the 732-utterance split did not appear on the test set. This is the attenuation observed across those five pairs, not a causal estimate of selection overfitting.

D.4 Language-model fusion

The 1.1 language model saw roughly 200k tokens, all drawn from transcripts used for acoustic fine-tuning, and changed WER by only 0.19 percentage points. The 1.3 language model saw 546k tokens, including 13.2k transcription lines unseen by the acoustic model, and reduced WER from 54.1% to 42.6% on the cleanly measured CTC comparison. Rescoring an autoregressive decoder's n-best list instead hurt, while beam search without the language model was also worse than greedy decoding.

D.5 Negative results and error analysis

Table 14 records the interventions that did not help, with the mechanism we attribute each failure to.

On the 1.3 validation split, code-switched utterances score 20.6% WER versus 22.1% for Arabic-

Intervention	Development result
MUSAN/RIR noise (1.1)	56.26 vs. 56.24 WER; no measurable gain.
Speed perturbation (1.3)	27.8 vs. 22.7; fast-speech buckets worsened.
Whisper full FT (1.1)	73.2; worse than every retained NeMo system.
LinTO acoustic data (1.3)	Negative at every tested mixture weight.
LM corpus scaling (1.3)	6.97M mixed tokens lost to 546k targeted tokens; selective line duplication distorted the count statistics used by Kneser–Ney smoothing (Kneser and Ney, 1995).
Self-training (1.3)	Agreement filtering retained shared model bias.
LLM correction (1.3)	Negative in three attempted configurations.

Table 14: Unsuccessful ASR interventions. These are development results, not official test scores.

only utterances; Arabic-token error is 20.5%, compared with 16.1% for Latin-script tokens. Normalizing recurrent Tunisian spelling variants recovers only 0.64 percentage points. WER is non-monotonic with speech rate: 24.5%, 18.9%, 21.4%, and 22.4% from the slowest to fastest buckets. On 1.1 development data, Mauritania and Algeria remain substantially harder than Egypt after adaptation (63.1% and 52.9% against 24.7%), motivating dialect-aware routing. The same ordering holds zero-shot, where Morocco (93.5%) and Mauritania (90.7%) trail Palestine (63.8%): adaptation compresses the gap without closing it.